R-LINEAR RIGHT EXACT DIRECT-SUM-COMMUTING FUNCTORS QCOH(X) → QCOH(Y) FOR AFFINE X ARE QCOH(X ×_R Y)

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file functors.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 0 hits for functors.tex (fu-occupied.json, 2026-10-14); consistent with the duty manager's six-check dispatch.

Source reference. Original equation 0FZC: sources/stacks-chow/tags.txt, source lines 19947–19947. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0FZB: sources/stacks-chow/tags.txt, source lines 19946–19946. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01SD: sources/stacks-chow/tags.txt, source lines 2334–2334. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01XC: sources/stacks-chow/tags.txt, source lines 2513–2513. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01SB: sources/stacks-chow/tags.txt, source lines 2332–2332. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01IB: sources/stacks-chow/tags.txt, source lines 1972–1972. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0GNQ: sources/stacks-chow/tags.txt, source lines 20800–20800. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let R be a ring. Let X and Y be schemes over R with X affine. There is an equivalence of categories between*

(1) *the category of R-linear functors $F : QCoh(\mathcal{O}_X) \to QCoh(\mathcal{O}_Y)$ which are right exact and commute with arbitrary direct sums, and*

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(2) *the category* $QCoh(\mathcal{O}_{X \times_R Y})$

given by sending $\mathcal{K}$ *to the functor* F *in (0FZC).*

Proof. Let $\mathcal{K}$ be an object of $QCoh(\mathcal{O}_{X \times_R Y})$ and $F_\mathcal{K}$ the functor (0FZC). By the discussion in Example 0FZB we already know that F is R-linear and commutes with arbitrary direct sums. Since $\mathrm{pr}_2 : X \times_R Y \to Y$ is affine (Morphisms, Lemma 01SD) the functor $\mathrm{pr}_{2,*}$ is exact, see Cohomology of Schemes, Lemma 01XC. Hence F is right exact as well, in other words F is as in (1).

Let F be as in (1). Say $X = \mathrm{Spec}(A)$. Consider the quasi-coherent $\mathcal{O}_Y$-module $\mathcal{G} = F(\mathcal{O}_X)$. The functor F induces an R-linear map $A \to \mathrm{End}_{\mathcal{O}_Y}(\mathcal{G})$, $a \mapsto F(a \cdot \mathrm{id})$. Thus $\mathcal{G}$ is a sheaf of modules over

$$A \otimes_R \mathcal{O}_Y = \mathrm{pr}_{2,*}\mathcal{O}_{X \times_R Y}$$

By Morphisms, Lemma 01SB we find that there is a unique quasi-coherent module $\mathcal{K}$ on $X \times_R Y$ such that $F(\mathcal{O}_X) = \mathcal{G} = \mathrm{pr}_{2,*}\mathcal{K}$ compatible with action of A and $\mathcal{O}_Y$. Denote $F_\mathcal{K}$ the functor given by (0FZC). There is an equivalence $\mathrm{Mod}_A \to QCoh(\mathcal{O}_X)$ sending A to $\mathcal{O}_X$, see Schemes, Lemma 01IB. Hence we find an isomorphism $F \cong F_\mathcal{K}$ by Lemma 0GNQ because we have an isomorphism $F(\mathcal{O}_X) \cong F_\mathcal{K}(\mathcal{O}_X)$ compatible with A-action by construction.

This shows that the functor sending $\mathcal{K}$ to $F_\mathcal{K}$ is essentially surjective. We omit the verification of fully faithfulness. □